Introduction to Full Stack Development & Web Architecture (MERN Overview)

1. Client (Browser)

- This is the application the user interacts with directly.
- Examples include Google Chrome, Mozilla Firefox, or Microsoft Edge.
- Displays the website or app interface
- Sends requests to the server (e.g., when you click a button or submit a form)
- Receives and renders responses (HTML, CSS, JavaScript)

2. Frontend (UI Layer)

- This is the part of the system that runs in the browser and controls what the user sees and interacts with.
- Built using technologies like HTML, CSS, and JavaScript
- Handles layout, design, and user interactions (buttons, forms, animations)
- Can use frameworks like React or Vue.js

3. Backend (Server)

- This is the “brain” of the system that processes requests from the frontend.
- Handles business logic (e.g., login validation, calculations)
- Receives requests from the client and sends back responses
- Communicates with the database
- Built using technologies like Node.js, Python, Java, etc.

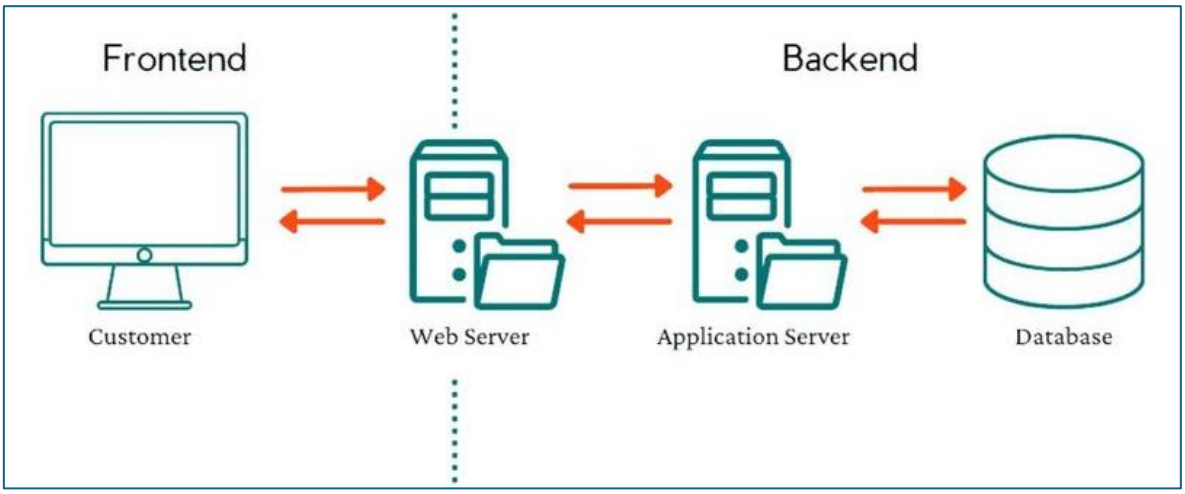
4. Database (MongoDB – Conceptual)

- This is where data is stored and retrieved.
- Stores user data, app content, records, etc.
- MongoDB is a NoSQL database that stores data in flexible, JSON-like documents
- The backend interacts with the database (not the frontend directly)

Simple Flow (How they work together):

1. User clicks something in the browser (Client)
2. Frontend sends a request to the Backend
3. Backend processes it and may query the Database
4. Database returns data to Backend
5. Backend sends response to Frontend
6. Frontend updates what the user sees

Sample Diagram:



MY PUBLIC GITHUB REPOSITORY:

https://github.com/GravityDefied4/fullstack-intro